Gated Position-Aided Fusion for Blockage-Robust Neural Beam Selection in 5G NR mmWave Systems

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***Abstract*—Beam selection in millimeter-wave (mmWave) 5G New Radio (NR) requires sweeping a large beam-pair codebook, incurring substantial initial-access overhead. A well-known neural approach predicts the full reference-signal-received-power (RSRP) profile over all beam pairs from a small set of downsampled RSRP measurements, enabling a top-K sweep over only the most promising beams. We first reproduce this regression benchmark on the standardized TR 38.901 urban macrocell scenario, attaining 90% top-K accuracy at $K = 13$ (81.4% sweep-overhead reduction). We then investigate whether fusing the already available user-equipment (UE) position with the sparse RSRP improves selection. Contrary to the common multi-modal intuition, we find that on *clean* RSRP the single-modality predictor is already near-optimal: position fusion, learned beam sampling, a beam-grid convolutional reconstruction, and uncertainty-aware adaptive-K all fail to beat it. The value of the position modality emerges only when the RSRP modality is *degraded*. Building on this, we propose a *gated residual position-aided fusion* network in which a position-driven correction is added to the RSRP-based estimate through a learned gate. The learned gate preserves clean-data accuracy at no cost—restoring full parity with the RSRP-only model, whereas an ungated residual forfeits roughly a point—while the residual path delivers a consistent top-K accuracy gain once measured beams are lost: $+2$ to $+3.2$ points (mean across five training seeds) under i.i.d. input dropout when 6–12 of the 14 measured beams are missing, rising to about $+7$ points under correlated (contiguous) loss, the regime closest to physical blockage. We stress that this is a measurement-availability stress test on the sparse RSRP input, not a channel-blockage simulation. An architecture ablation confirms the gate is responsible for this no-regret behavior. The result is a practical robustness improvement for blockage-prone mmWave links at no clean-data cost.**

***Index Terms*—5G NR, millimeter wave, beam management, beam selection, deep learning, multi-modal fusion, beam blockage, RSRP.**

I. Introduction

Millimeter-wave (mmWave) bands are central to 5G NR and 6G for their large bandwidths, but their high pathloss and susceptibility to blockage demand highly directional beamforming [1]. During initial access, the gNB and UE must establish a good transmit/receive beam pair, classically through an exhaustive synchronization-signal-block (SSB) beam sweep over the full codebook. With many narrow beams, this sweep dominates initial-access latency and overhead, motivating machine-learning methods that predict good beams from partial or out-of-band observations [2], [3].

Two broad families have emerged. *In-band* methods exploit RSRP or channel measurements [4], [5]; they are accurate when measurements are reliable. *Side-information* methods exploit position, LiDAR, radar, vision, or sub-6 GHz channels [6]–[10]; they are robust to RF impairments but typically achieve lower peak accuracy. A widely used in-band formulation, distributed as a reference example by MathWorks [11],[1] casts beam selection as RSRP-profile *regression*: a neural network maps a downsampled subset of RSRP measurements to the full per-beam-pair RSRP profile, after which only the top-K predicted beams are swept. This example was *recently updated* from an earlier position-input classifier to the regression formulation we reproduce; we reconcile the two versions in Section III.

In this work we take this regression benchmark as our starting point and ask a simple question: *does adding the UE position—already available at the network—improve beam selection?* Our investigation yields a nuanced and, we believe, instructive answer. The contributions are:

- We reproduce the SSB-RSRP regression benchmark on the TR 38.901 urban macrocell (UMa) scenario and confirm its headline operating point: 90% top-K accuracy at $K = 13$, i.e. an 81.4% reduction in beam-sweep overhead, with average RSRP within 0.24 dB of exhaustive search (Section III).
- We show that, on clean RSRP, the single-modality predictor is at the achievable ceiling: naive position fusion, statistically "optimized" beam sampling, a 2D beam-grid convolutional reconstruction, and uncertainty-aware adaptive-K all fail to improve it (Section IV). This isolates *where* a second modality can help.
- We propose a *gated residual position-aided fusion* network with an impairment-aware training procedure (Section V). Its central component is a learned gate whose role is *clean-data no-regret parity*: trained end-to-end with no manual tuning, it restores clean-data parity (gated 89.4%, matching the RSRP-only reference) that an ungated residual forfeits (88.0%), so the position

[1] This is the project's designated benchmark (Option B, *Neural Network for Beam Selection*).

Both authors contributed equally to this work.

modality can only help and never hurt. (By *no-regret* we mean only that adding the position modality incurs no clean-data accuracy penalty relative to the RSRP-only model—not the online-learning notion of regret.) On top of this parity, the residual position path improves top-K accuracy by $+2$ to $+3.2$ points (mean across five seeds, each gain exceeding twice its across-seed std) under i.i.d. measurement loss, rising to about $+7$ points under correlated (contiguous) loss (Section VI). As a finding, the learned gate converges to a near-constant scale: in this single-impairment regime (one always-reliable position source) a fixed-scale residual suffices and is no-regret, which in turn motivates genuinely adaptive gating only when position reliability *itself* varies—a direction we outline in Section VII.

The complete MATLAB implementation, trained models, and the LaTeX source of this paper are publicly available on GitHub,[2] and the paper source is also shared as an Overleaf project.[3]

II. Related Work (Literature Review)

Early ML beam-selection work posed the task as classification of the best beam (pair) from side information such as UE position or LiDAR [2], [3], [6], a classification line that continues on real V2I data [12]. Position/location-aided prediction was extended to 3D beam selection with orientation information [7], and to real-world GPS-based prediction on the DeepSense 6G testbed [8]. Out-of-band approaches predict mmWave beams (and blockages) from sub-6 GHz channels [4], [10], [13], reporting large overhead reductions (e.g. $\sim$79% in [4]). Sensing-aided prediction using radar [9] and LiDAR [14] has been demonstrated on real V2I data, and waveform-learning approaches such as DeepBeam [5] reduce beam-management latency without coordination, while entropy-based probing-beam selection cuts measurement overhead [15]. A parallel line predicts *blockage* for proactive handoff [16], [17], and recent work exploits temporal correlation for vehicular beam management [18]. Surveys [1] and the 3GPP study on AI/ML for the NR air interface [19] provide broader context.

Table I summarizes representative methods by input modality and reported benefit; it is grouped by period so that the eight recent (2021–2025) works defining the current state of the art appear above the foundational pre-2021 methods. Two gaps motivate our work. First, multi-modal fusion is typically studied with rich external sensors (camera/LiDAR/radar) on vehicular data, not with the *already-available* UE position fused into the canonical SSB-RSRP regression. Second, prior fusion work emphasizes peak accuracy on nominal data rather than *robustness as a function of measurement reliability*. We address both, and—importantly—report the negative result that on clean RSRP the position modality is redundant, which clarifies that the fusion payoff is a robustness, not an accuracy, story.

[2]Code and data: https://github.com/fyakkan/neural-beam-selection

[3]Overleaf project: https://www.overleaf.com/read/ydgmcckxxzkm#8a1c66

III. System Model and Regression Benchmark

A. Scenario and beam codebook

We adopt the TR 38.901 UMa scenario [20] in frequency range 2 (FR2) at a carrier of 30 GHz, following the AI/ML system-level assumptions of TR 38.843 [19]. The gNB uses a 4×8 cross-polarized panel and the UE a 1×4 panel, yielding $N_{\text{tx}} = 10$ transmit beams and $N_{\text{rx}} = 7$ receive beams, for a total of $N_b = N_{\text{tx}}N_{\text{rx}} = 70$ beam pairs covering a 120° sector. For each UE location, SSB-based beam sweeping yields an RSRP value for every beam pair, forming an RSRP matrix $\mathbf{R} \in \mathbb{R}^{7\times 10}$; we index beam pairs by the column-major linear index $b \in \{1,\ldots,70\}$ of $\mathbf{R}$. We use the official pre-recorded dataset of 15,000 training and 500 test UE locations. Blocked beams appear as $-\infty$ RSRP (4.6% of entries; $\sim$10% of locations have at least one blocked beam among the sampled set); we floor them to -120 dB for network inputs/targets.

B. Regression formulation

Let $\mathbf{r} \in \mathbb{R}^{70}$ be the vectorized RSRP profile of a UE. A fixed downsampling selects every fifth beam pair, giving $\mathcal{S} = \{1, 6, 11, \ldots, 66\}$ with $|\mathcal{S}| = 14$ measured beams and input $\mathbf{x} = \mathbf{r}_{\mathcal{S}} \in \mathbb{R}^{14}$. A network $f_\theta : \mathbb{R}^{14} \to \mathbb{R}^{70}$ regresses the full profile $\hat{\mathbf{r}} = f_\theta(\mathbf{x})$; targets are min-max normalized to $[-1, 1]$ and the output layer is $\tanh$. At inference, the K beams with the largest predicted RSRP are swept and the one with the strongest *actual* RSRP is selected. The metrics are the top-K accuracy (the fraction of UEs whose true-optimal beam is among the K predicted) and the average selected RSRP.

The benchmark network is a four-hidden-layer multilayer perceptron (MLP) of width 96 with leaky-ReLU activations and z-score input normalization, trained with Adam to minimize mean-squared error. Position is *not* an input to the network; it is used only by a K-nearest-neighbor (KNN) benchmark, alongside Statistical and Random baselines.

C. Reproduction results

Table II reports the reproduced metrics and Fig. 1 the top-K accuracy curves. The neural predictor reaches 90.0% top-K accuracy at $K = 13$—sweeping fewer than one quarter of the 70 beam pairs, an 81.4% overhead reduction—and 97.4% at $K = 30$. At the low-K points listed in the project brief it attains 26.0%, 43.4%, 55.2%, and 72.4% for $K = 1, 2, 3, 5$ (Table II), each far above the position-only KNN. Its average RSRP at $K = 13$ is -23.21 dB, within 0.24 dB of the exhaustive optimum (-22.97 dB) and well above KNN (-25.29 dB). The neural method dominates KNN, Statistical, and Random across all K, reproducing the published behavior and validating our pipeline.

Reconciliation with the original formulation. The project brief describes Option B as a position-input multi-class *classifier*—predicting the optimal beam-pair index directly from the UE's 3D coordinates—and this is exactly the variant that ships with our locally installed MATLAB R2025b. The publicly documented (online) MathWorks example [11], however, has since been updated to the RSRP-profile *regression* formulation reproduced above, and is the live reference today. We adopt

TABLE I

REPRESENTATIVE ML BEAM-MANAGEMENT METHODS, GROUPED BY PERIOD: THE EIGHT RECENT (2021–2025) STATE-OF-THE-ART WORKS ARE LISTED FIRST, ABOVE THE FOUNDATIONAL PRE-2021 METHODS.

Work (year)	**Modality**	**Approach**	**Reported benefit**
Recent state of the art (2021–2025)			
Polese *et al.* [5] (2021)	Waveform/RF	Coordination-free beam mgmt	$\sim$7$\times$ lower latency (default config)
Rezaie *et al.* [7] (2022)	Position + orientation	DL 3D beam set selection	Location/orientation-aided selection
Demirhan & Alkhateeb [9] (2022)	Radar	DL beam prediction	Real-world radar-aided prediction
Wu *et al.* [17] (2022)	In-band power	Moving-blockage prediction	>85% blockage accuracy
Jiang *et al.* [14] (2023)	LiDAR	DL future-beam prediction	$\sim$90% overhead reduction (V2I)
Biliaminu *et al.* [12] (2024)	Position/geo	ML multiclass beam classification	Beam prediction on real V2I (DeepSense)
Meng *et al.* [15] (2024)	RSRP/probing	Entropy-based probing + DL prediction	Reduced probing/measurement overhead
Oliveira *et al.* [18] (2025)	Position + time	RNN temporal beam mgmt	Up to 50% overhead reduction
Foundational methods (pre-2021)			
Heng & Andrews [2] (2019)	Position	NN beam alignment	ML-assisted alignment vs. exhaustive sweep
Klautau *et al.* [6] (2019)	LiDAR	DL beam selection	Out-of-band beam selection
Sim *et al.* [4] (2020)	Sub-6 GHz	DL beam selection (NR/6G)	$\sim$79% sweep-overhead reduction
Alrabeiah & Alkhateeb [13] (2020)	Sub-6 GHz	DL beam + blockage prediction	>90% blockage prediction
This work	**Sparse RSRP + position**	**Gated residual fusion**	**No clean-data cost; $+2$ to $+3.2$ pts (i.i.d.), up to $\sim$$+7$ (contiguous)**

this current (regression) version for two reasons: it is the maintained reference, and it cleanly separates the in-band (RSRP) and side-information (position) modalities, which is exactly what our fusion study requires (position becomes an *added* modality rather than the sole input). We confirmed with the course instructor that reproducing the current regression formulation satisfies the benchmark-reproduction requirement. We re-implement rather than re-run the shipped code: the online example uses a wider 64–128–256–128 ReLU network with global-max input normalization on a $20{,}000/700$ split, whereas we use a compact width-96 leaky-ReLU z-score MLP on the $15{,}000/500$ pre-recorded dataset bundled with the example. Despite these differences, our re-implementation reproduces the published operating point *exactly* (90.0% top-K accuracy at $K = 13$, an 81.4% sweep-overhead reduction), confirming functional equivalence; the RSRP-only reference used throughout our experiments shares this width-96 architecture, so all comparisons are internally fair.

IV. WHERE A SECOND MODALITY DOES (NOT) HELP

Before designing a fusion model, we tested whether the benchmark can be improved on *clean* RSRP along four axes, each under identical data, splits, and seeds:

1) **Naive position fusion**: a two-branch network ingesting the 14 RSRP values and the 3D UE position. Result: a tie within noise on clean data ($+0.8$ at K=13, -3.0 at K=5).
2) **Learned/statistical beam sampling**: replacing the uniform every-fifth sampling with the 14 highest-variance, most-frequently-optimal, or strongest-mean beams. All were *worse* (e.g. K_{90}=39 vs. 13), because clustered beams lose spatial coverage; uniform sampling is near-optimal.
3) **2D beam-grid CNN**: treating sparse sensing as inpainting on the 7×10 (Rx$\times$Tx) RSRP image with observed-value and mask channels. Result: worse at the operating point (K_{90}=16 vs. 13).
4) **Uncertainty-aware adaptive-K**: a nucleus rule over the predicted profile to choose a per-UE K. Result: negligible ($\leq +0.6$ pts at matched mean overhead).

TABLE II

REPRODUCED BENCHMARK METRICS ON THE $15{,}000/500$ SPLIT. TOP-K ACCURACY IS LISTED AT THE PROJECT BRIEF'S K=1, 2, 3, 5 POINTS AND AT THE K=13 OPERATING POINT (WHERE THE PREDICTOR FIRST REACHES 90%); OTHER ROWS ARE AT K=13 UNLESS NOTED.

Metric	Neural	KNN	Optimal
Top-K accuracy @ K=1	26.0%	4.6%	–
Top-K accuracy @ K=2	43.4%	10.4%	–
Top-K accuracy @ K=3	55.2%	14.2%	–
Top-K accuracy @ K=5	72.4%	21.8%	–
Top-K accuracy @ K=13	**90.0%**	37.8%	–
Top-K accuracy @ K=30	97.4%	–	–
Overhead reduction @ K=13	**81.4%**	–	–
Avg. RSRP @ K=13 (dB)	-23.21	-25.29	-22.97

The four comparisons above use the *same* data, train/validation/test split, and random seed as the RSRP-only reference, so each delta is an apples-to-apples single-run point estimate. We deliberately treat differences within the $\sim$1-point run-to-run noise that we measure for these compact networks (cf. the per-draw and per-seed standard deviations in Tables III–V) as ties, not improvements; the gross failures

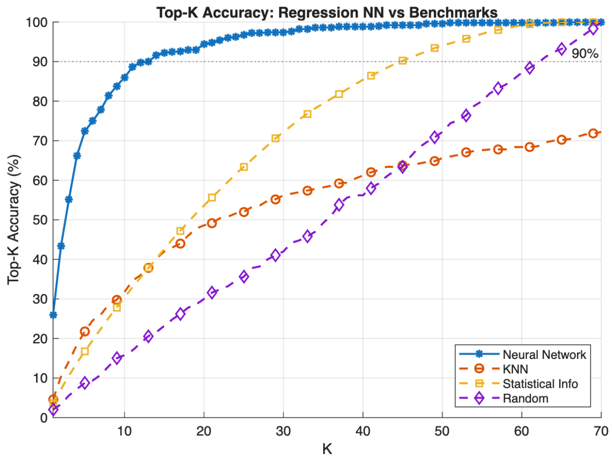

Fig. 1. Reproduced benchmark: top-K accuracy of the RSRP-regression network vs. KNN, Statistical, and Random baselines. The network reaches 90% at $K = 13$.

we cite (K_{90}=39 vs. 13 for "smart" sampling, K_{90}=16 vs. 13 for the CNN) lie far outside that band and are not artifacts of seed choice. The common explanation is that the optimal beam is determined by the channel, which the RSRP directly measures; the UE position predicts the optimal beam only weakly. This is not a weak baseline by construction but an intrinsic property of the UMa scenario: the optimal beam pair is set by the instantaneous multipath channel (scattering geometry, UE orientation, small-scale fading), so UEs at nearly identical 3D positions routinely have different optimal beams. Position is therefore a coarse, many-to-one predictor of the best beam, which caps any position-only method—the KNN benchmark attains only 37.8% top-K at K=13 not because it is poorly tuned but because position alone does not determine the beam. Hence, when RSRP is clean, position is largely *redundant*, and richer architectures merely add variance. This motivates targeting the regime where RSRP is unreliable, where the (weak but RF-impairment-immune) position signal becomes complementary rather than redundant.

V. Proposed Method: Gated Residual Position-Aided Fusion

A. Beam-blockage model

Practical mmWave sweeps lose measurements to blockage and missed detections. We model this by, for each UE, setting n_B of the 14 measured beams (chosen uniformly at random) to the floor value, plus a small Gaussian RSRP measurement noise. The number of blocked input beams n_B parameterizes the impairment severity; n_B large corresponds to only a handful of reliably measured beams. *Scope of the model.* It is important to be precise about what this impairment represents: it drops n_B of the 14 *measured input* beams while leaving the regression *target*—the full 70-beam RSRP profile and hence the true optimal beam—unchanged. It is therefore a measurement-availability stress test on the sparse network input, not a physical channel-blockage simulation (which would degrade the actual channel of the occluded directions, possibly including the optimal beam itself, and change the ground-truth label). This isolates exactly the question we study—how the two modalities cope when part of the in-band measurement is unavailable—and we qualify the term "blockage" in this sense throughout. We note that this model drops beams independently and uniformly at random, whereas physical mmWave blockage is spatially *correlated*—a single obstruction tends to occlude a contiguous set of beam directions. We adopt the i.i.d. model as our primary setting and separately verify robustness under correlated (contiguous) blockage in Section VI-E.

B. Architecture

The proposed network has two encoders. An RSRP encoder maps $\mathbf{x}$ to a primary estimate $\mathbf{b} = g_{\text{base}}(\mathbf{x}) \in \mathbb{R}^{70}$ (structurally the RSRP-only path). A position encoder maps the 3D UE position $\mathbf{p}$ to a correction $\mathbf{c} = g_{\text{corr}}(\mathbf{p}) \in \mathbb{R}^{70}$. A gate $\mathbf{g} = \sigma(g_{\text{gate}}([\mathbf{h}_r, \mathbf{h}_p])) \in (0,1)^{70}$, computed from both encoder features $\mathbf{h}_r, \mathbf{h}_p$, weights the correction. The output is

$$\hat{\mathbf{r}} = \tanh\big(\mathbf{b} + \mathbf{g} \odot \mathbf{c}\big), \tag{1}$$

where $\odot$ is the elementwise product. The residual-plus-gate structure lets the network recover the strong RSRP-only estimate (by suppressing $\mathbf{g} \odot \mathbf{c}$) when RSRP is reliable, and inject the position prior when it is not. The RSRP encoder uses two width-96 layers, the position encoder two width-32 layers, and the gate a width-64 layer; all use leaky-ReLU. The position and gate branches are deliberately small, so the gated network's parameter count and inference latency are comparable to the RSRP-only reference; the only added cost is reading the already-available UE position and one extra forward pass. Our use of "no-regret" (defined in the introduction) therefore refers to clean-data *accuracy* parity, with this minor compute overhead acknowledged rather than claimed away.

C. Impairment-aware training

Both the proposed model and the RSRP-only reference are trained with the *same* blockage augmentation: per sample, $n_B \sim \mathcal{U}\{0, \ldots, 8\}$ beams are dropped and light noise is added. This teaches the models to operate across blockage levels and makes the comparison differ only in the position modality. Networks are trained with Adam (300 epochs, mini-batch 512, initial learning rate 10^{-3} with piecewise decay).

VI. Experimental Results

A. Experimental setup

All experiments use the TR 38.901 UMa scenario at 30 GHz (FR2), with a 4×8 gNB panel and 1×4 UE panel giving 70 beam pairs; the network input is the 14 uniformly downsampled RSRP measurements. We use the official pre-recorded dataset of 15,000 training and 500 test UE locations, holding out 10% of training data for validation (13,500/1,500/500 train/validation/test). All networks are trained with Adam for 300 epochs, mini-batch 512, initial learning rate 10^{-3} with piecewise decay ($\times 0.5$ every 80 epochs), minimizing mean-squared error on the $[-1, 1]$-normalized RSRP profile; the proposed and RSRP-only models additionally use blockage augmentation ($n_B \sim \mathcal{U}\{0, \ldots, 8\}$). For the headline blockage-robustness comparison (Table III) we retrain both models across *five* random seeds and report mean$\pm$std across seeds, with each per-seed value averaged over 10 independent blockage draws; this characterizes training-initialization variance in addition to test-time draw variance. The architecture ablation (Table V) and the auxiliary studies are reported at a single representative seed (seed 1) over 10 draws so that all variants share identical draws. Experiments run in MATLAB R2025b (Deep Learning, 5G, Communications, Phased Array System, Statistics and Machine Learning, DSP System, and Signal Processing Toolboxes); the networks are compact enough to train on CPU in a few minutes each (no GPU required).

B. Robustness to beam blockage

Recall that the clean benchmark already cuts beam-sweep overhead by 81.4% at 90% top-K accuracy (K=13; Section III); the question here is whether that operating point survives lost input measurements, and whether the position modality helps preserve it. Fig. 2 and Table III report top-K accuracy ($K = 13$) versus the number of blocked input beams. To ensure the comparison is not an artifact of a single weight initialization, we retrained *both* models across five random seeds and report every cell as mean$\pm$std *across seeds* (each per-seed value is itself averaged over 10 i.i.d. blockage draws); the across-seed std thus captures training-initialization and augmentation variance, which a per-draw std does not. On clean inputs ($n_B = 0$) the proposed fusion matches the RSRP-only model within seed noise (89.0% vs. 88.7%; gain $+0.3 \pm 0.6$): the position modality incurs *no* clean-data cost, and— notably—the fusion is the *more stable* model (across-seed std 0.4 vs. 0.8).[4] As beams are blocked, the fusion pulls ahead, with a gain that grows to $+2.5 \pm 0.7$, $+2.3 \pm 0.8$, and $+3.2 \pm 0.6$ points at $n_B = 6, 8, 10$. The position-only KNN is blockage-immune but flat at 37.8%, far below both networks except under extreme blockage. Fig. 3 shows that, at a representative $n_B = 8$, the fusion dominates the RSRP-only model across the entire K range. The seed-level view sharpens significance (Table III): the differences at low blockage ($n_B \leq 4$: $+0.3$, -1.0, $+0.8$) are small and not consistently signed, lying within roughly one to two across-seed standard deviations of zero, whereas the gains for $n_B \in [6, 12]$ ($+2.3$ to $+3.2$) all exceed *twice* their across-seed std and constitute the real effect that survives initialization noise. Note that the models are trained with $n_B \sim \mathcal{U}\{0, \ldots, 8\}$, so the results at $n_B \in \{10, 12\}$ additionally probe generalization *beyond* the trained blockage range, where the fusion advantage is largest.

[4]The clean (n_B=0) accuracies here (across-seed means 88.7% RSRP-only, 89.0% fusion) sit about a point below the 90.0% of Section III because these models are trained with blockage augmentation, which costs $\sim$1 point on clean inputs; the width sweep in Section VI likewise reports 89.6% for the narrowest (width-32) network.

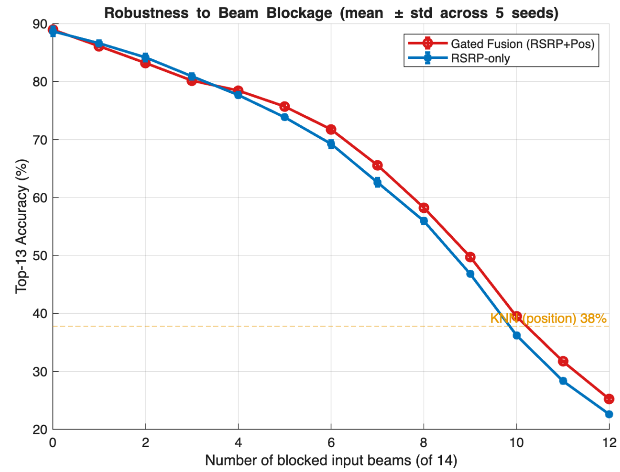

Fig. 2. Top-K accuracy (K=13) vs. number of blocked input beams; error bars are ± 1 standard deviation *across 5 random training seeds* (each point averaged over 10 i.i.d. blockage draws). The gated fusion matches RSRP-only when clean—and is more stable—and improves it under blockage; KNN (position only) is flat.

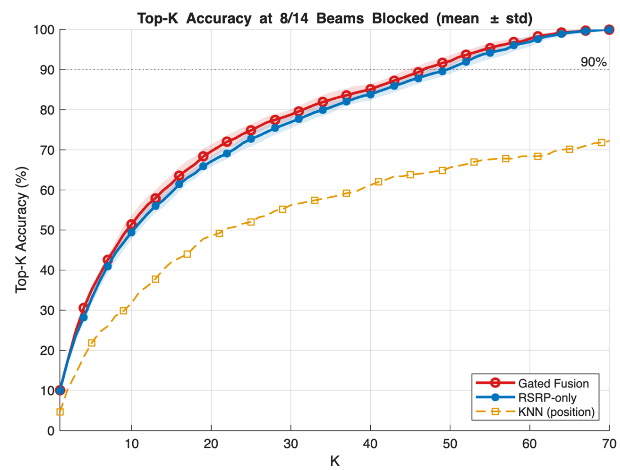

Fig. 3. Top-K accuracy vs. K at a representative 8/14 blocked input beams; shaded bands are ± 1 standard deviation over 10 draws. The gated fusion dominates the RSRP-only model across K; both far exceed position-only KNN.

Link quality, not just hit rate. Top-K accuracy is a selection proxy; the deployable quantity is the quality of the link actually established. We therefore also report, in Table IV, the average RSRP of the beam *selected* after the top-K=13 sweep under i.i.d. input loss. The fusion's accuracy advantage translates directly into a better link: at n_B=8 it attains -24.52 dB versus the RSRP-only model's -25.08 dB ($+0.56$ dB), and the margin widens to $+1.26$ dB at n_B=10 as more beams are lost. The gain is thus a genuine link-budget improvement, not a metric artifact, and—like the accuracy gain—it grows with impairment severity.

C. Architecture ablation: is the gate needed?

Table V compares four fusion variants trained identically: RSRP-only, concatenation fusion, plain residual ($\mathbf{g} = 1$ in (1)), and the proposed gated residual. Because cleanly isolating the

TABLE III

TOP-K ACCURACY (K=13, %) VS. NUMBER OF BLOCKED INPUT BEAMS, REPORTED AS MEAN±STD *across 5 random training seeds* (EACH PER-SEED VALUE IS ITSELF AVERAGED OVER 10 I.I.D. BLOCKAGE DRAWS). THE ACROSS-SEED STD THEREFORE CAPTURES TRAINING-INITIALIZATION AND AUGMENTATION-STREAM VARIANCE—WHICH A PER-DRAW STD DOES NOT—SO THE GAINS ARE SHOWN TO SURVIVE THE CHOICE OF SEED. THE GAIN ROW IS THE MEAN OF PER-SEED (FUSION−RSRP-ONLY) DIFFERENCES; SINCE EACH ROW IS ROUNDED INDEPENDENTLY, A GAIN CELL CAN DIFFER FROM THE ROUNDED ROW DIFFERENCE BY 0.1 (E.G. AT n_B=10).

# blocked	0	2	4	6	8	10	12
RSRP-only	$88.7_{\pm 0.8}$	$84.2_{\pm 0.6}$	$77.7_{\pm 0.5}$	$69.3_{\pm 0.6}$	$56.0_{\pm 0.4}$	$36.2_{\pm 0.4}$	$22.6_{\pm 0.3}$
Gated fusion	$89.0_{\pm 0.4}$	$83.2_{\pm 0.3}$	$78.5_{\pm 0.5}$	$71.8_{\pm 0.3}$	$58.3_{\pm 0.4}$	$39.5_{\pm 0.3}$	$25.2_{\pm 0.3}$
Gain (pts)	$+0.3_{\pm 0.6}$	$-1.0_{\pm 0.7}$	$+0.8_{\pm 0.7}$	$+2.5_{\pm 0.7}$	$+2.3_{\pm 0.8}$	$+3.2_{\pm 0.6}$	$+2.6_{\pm 0.3}$

TABLE IV

AVERAGE RSRP (DB) OF THE *selected* BEAM AFTER THE TOP-K=13 SWEEP VS. NUMBER OF BLOCKED INPUT BEAMS (I.I.D., MEAN OVER 10 DRAWS). HIGHER IS BETTER; THE EXHAUSTIVE OPTIMUM IS −22.97 DB. GAINS ARE COMPUTED FROM UNROUNDED VALUES.

# blocked	0	6	8	10	12
RSRP-only	−23.23	−24.11	−25.08	−27.20	−28.61
Gated fusion	−23.21	−23.84	−24.52	−25.95	−27.45
Gain (dB)	+0.01	+0.27	+0.56	+1.26	+1.16

TABLE V

ARCHITECTURE ABLATION (SINGLE REPRESENTATIVE SEED, SEED 1): TOP-K ACCURACY (K=13, %, MEAN±STD OVER 10 BLOCKAGE DRAWS) VS. BLOCKAGE.

# blocked	0	6	8	10	12
RSRP-only	$89.2_{\pm 0.0}$	$69.8_{\pm 1.5}$	$56.0_{\pm 1.6}$	$35.9_{\pm 2.4}$	$22.3_{\pm 0.9}$
Concatenation	$84.4_{\pm 0.0}$	$69.7_{\pm 1.7}$	$59.8_{\pm 1.5}$	$45.9_{\pm 1.2}$	$29.1_{\pm 1.2}$
Plain residual	$88.0_{\pm 0.0}$	$71.7_{\pm 1.8}$	$58.0_{\pm 2.1}$	$39.7_{\pm 1.8}$	$25.0_{\pm 1.4}$
Gated residual	$89.4_{\pm 0.0}$	$72.1_{\pm 1.7}$	$57.9_{\pm 2.0}$	$39.7_{\pm 1.7}$	$24.9_{\pm 1.4}$

gate requires all four architectures to see *identical* blockage draws, this ablation is reported for a single representative seed (seed 1); its RSRP-only and gated columns are the seed-1 instances, whose blockage gains lie within the across-seed band of Table III. The plain residual already provides most of the blockage robustness but costs 1.2 points on clean data (88.0% vs. 89.2%): an always-on position correction slightly perturbs the near-optimal clean RSRP estimate. The gate restores clean parity (89.4%) at equal blockage robustness—this clean-data no-regret behavior is precisely the contribution of the gate, and it is what makes the position modality strictly safe to add. Interestingly, ungated *concatenation* achieves the largest heavy-blockage gains (45.9% vs. 35.9% at $n_B = 10$) but sacrifices 4.8 points on clean data, i.e. it is *not* no-regret. The gated residual occupies the best clean-vs-robustness trade-off. Notably, the learned gate converges to a nearly constant value (≈ 0.78; Fig. 4): rather than "opening" adaptively with blockage, it learns a fixed down-weighting of the position correction that preserves clean accuracy while retaining robustness. We therefore frame the gate's role as learning the (near-constant) optimal position-mixing scale rather than performing per-sample adaptation. Two points keep this a *finding* rather than a deflation. First, the scale is discovered automatically by end-to-end training—not hand-set—so the method needs no per-deployment grid search; the gate is what *lets* training find the no-regret operating point that the ungated residual misses. Second, that the optimum is near-constant is an empirical property of *this* regime (a single, always-reliable position source), not a general claim: it precisely predicts when adaptivity is unnecessary, and motivates genuinely adaptive gating only when position reliability *itself* varies across users (e.g. GPS outages), which we leave to future work (Section VII).

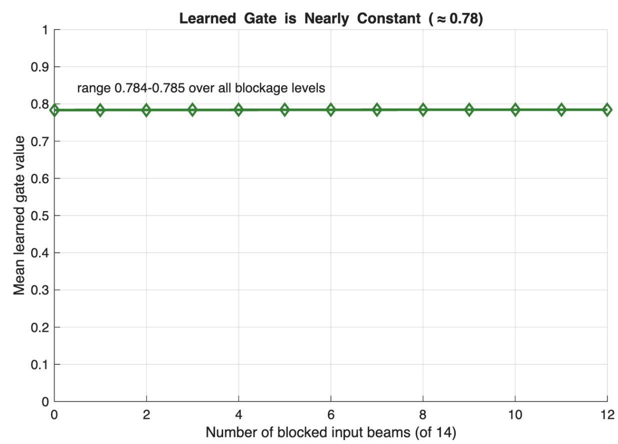

Fig. 4. The learned gate is nearly constant (≈ 0.78) across blockage levels: robustness arises from the always-on residual position path, not from adaptive gating.

D. Benchmark sensitivity (further analyses)

We characterize the RSRP-only benchmark on clean data. *Width*: increasing the hidden width from 32 to 384 raises top-K accuracy modestly (89.6% to 91.8% at K=13); width 96 is a good accuracy/size balance. *Depth*: 2–4 hidden layers perform equally ($\approx 90\%$), while 6–8 layers degrade slightly (mild overfitting), justifying the four-layer choice. *UE density*: accuracy saturates by $\sim 25\%$ of the training set (3,375 UEs, 90.6%), so the full 15k set is comfortably in the saturated regime—confirming that the clean-data ceiling is not a data-limited artifact.

E. Robustness to correlated (contiguous) blockage

To probe a more realistic impairment, we additionally evaluate *correlated* blockage: rather than dropping the n_B lost beams i.i.d., we drop a *contiguous* block of n_B sampled

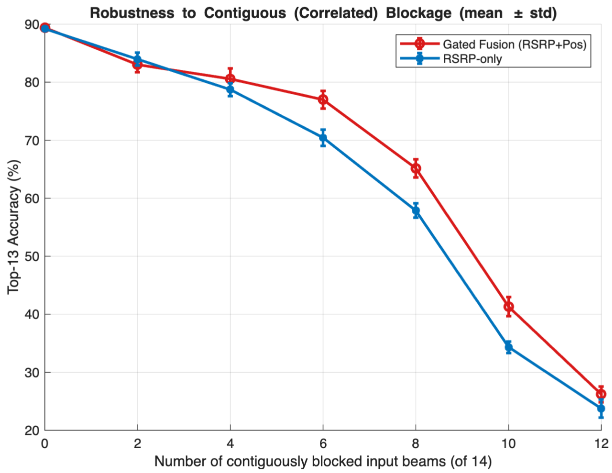

Fig. 5. Additional robustness study: top-K accuracy (K=13) under *contiguous* (correlated) blockage of the sampled beams; error bars are ± 1 std over 10 draws. The gated-fusion advantage over RSRP-only is larger than under i.i.d. blockage (cf. Fig. 2).

beams (a single obstruction occluding a contiguous angular region), reusing the same i.i.d.-trained models (the representative seed 1, for a like-for-like comparison)—i.e. a test-time distribution shift. Fig. 5 shows that the gated-fusion advantage not only persists but *grows*: the gain rises to $+6.5$, $+7.3$, and $+7.0$ points at $n_B = 6, 8, 10$, versus $+2.3$, $+2.0$, $+3.8$ for the *same* model under i.i.d. blockage (consistent with the across-seed $+2.5$, $+2.3$, $+3.2$ of Table III), well beyond the ≤ 2-point per-draw standard deviation. Intuitively, a contiguous gap removes an entire local region of the RSRP profile that cannot be interpolated from neighboring samples, so the global position prior contributes more. The method is thus, if anything, *more* valuable under the structured blockage expected in practice.

F. Discussion

Our results delineate the operating regime of multi-modal beam selection. When RSRP is clean and densely informative, single-modality regression is essentially optimal and additional modalities or architectures provide no gain—a useful negative result that tempers the multi-modal intuition for this setting. The position modality becomes valuable only as the RSRP modality degrades, exactly the condition created when mmWave input measurements become unavailable—the measurement-dropout regime of Section V, induced in practice by blockage and missed detections. The gated residual design exploits this asymmetry: it defaults to the RSRP path and adds a bounded position correction, yielding a no-regret method that is strictly preferable to deploying either a position-blind predictor or an always-on (concatenation) fusion that taxes clean-data accuracy. The trade-off exposed in Table V also offers a design knob: deployments dominated by blockage may prefer concatenation for its larger heavy-blockage gains, accepting a clean-data penalty.

G. Positioning against the state of the art

A direct numerical ranking against the works in Table I is confounded by differing scenarios, datasets, and codebooks (e.g. real-world V2X with cameras/LiDAR/radar [8], [9], [14] vs. our TR 38.901 SSB-RSRP setting), so we contextualize rather than rank. Side-information methods report strong overhead reductions—$\sim$79% from sub-6 GHz [4] and $\sim$90% from LiDAR in V2I [14]—but rely on rich external sensors and largely vehicular data, while position-only prediction [7], [12] trades peak accuracy for robustness. Our reproduced benchmark already attains an 81.4% overhead reduction at 90% top-K from in-band RSRP alone; our contribution is *orthogonal* to the accuracy-focused fusion literature: a $+2$ to $+7$-point accuracy retention under beam blockage at *no* clean-data cost, using only the already-available UE position rather than added sensors. To our knowledge, this no-regret, reliability-oriented fusion of position with sparse RSRP—and the explicit characterization of *when* a second modality helps—is not covered by the surveyed works.

VII. Conclusion

We reproduced the SSB-RSRP regression benchmark for 5G NR mmWave beam selection (90% top-K accuracy at $K = 13$, an 81.4% sweep-overhead reduction) and studied position-aided multi-modal fusion. We found that on clean RSRP the single-modality predictor is at the achievable ceiling, but that a gated residual fusion of sparse RSRP and UE position yields a no-regret robustness improvement: clean-data parity—the learned gate restoring the $\sim$1 point an ungated residual would forfeit—together with a consistent top-K accuracy gain when input measurements are lost ($+2$ to $+3.2$ points, mean across five seeds, under i.i.d. dropout, rising to about $+7$ points under correlated/contiguous loss) and a matching improvement in selected-beam RSRP (up to $+1.3$ dB). An ablation attributes the no-regret property to the learned gate, and shows the gate converges to a near-constant mixing scale in this single-reliability regime. *Limitations.* Our impairment is input measurement dropout on the sparse RSRP input, not a physical channel-blockage simulation (which would also alter the ground-truth optimal beam), and we use the single, always-reliable UE position of the official dataset; the gains are therefore reported on this controlled stress test rather than on field blockage traces. Future work includes joint optimization of the measured-beam subset with the fusion network, evaluation on real multi-modal datasets such as DeepSense 6G, and—since the learned gate is near-constant when position is always reliable—studying *variable position reliability* (e.g. GPS noise/outage), where an adaptive gate that suppresses the position path per-user is expected to provide gains beyond the fixed-scale residual reported here.

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